

Madhusudan Ambulkar

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SKILLS

BI & Analytics : Power BI (DAX, Time Intelligence, KPI Dashboards), SQL (joins, CTEs, window functions), MySQL, Snowflake, DBeaver, data mining, ETL pipelines, schema validation

Programming & Tools : Python (Pandas, NumPy, scikit-learn), Excel (VBA/Macros), Jupyter Notebook, Git/GitHub, VS Code, Linux/Shell

AI & ML : YOLOv8 (object detection, edge deployment), LLM APIs (Claude, OpenAI), Prompt Engineering, Power Automate, AI agent concepts (MCP, RAG, LangChain)

Dashboards & Reporting : Streamlit (hands-on), Power BI, Google Looker Studio, stakeholder-facing dashboards, SOP writing

Cloud & DevOps : Microsoft Azure, GCP (BigQuery basics), Docker (basics), Atlassian JIRA, Agile workflows

Languages : English (C1 – IELTS 7.0), German (A1.2, actively improving)

EXPERIENCE

eClinicalWorks

Jul 2022 – Aug 2024

Software Specialist

Mumbai, India

- Investigated and resolved 300+ data and system issues using SQL queries in DBeaver across MySQL and Snowflake databases, improving team resolution efficiency by 20% within an agile sprint environment.
- Designed and maintained data visualisations in Power BI (DAX, time intelligence) and Excel, translating complex operational data into clear dashboards for non-technical stakeholders across sales and product teams.
- Documented and re-engineered 10+ internal SOPs for sales and operations workflows, cutting incident escalation time by 15% and reducing onboarding friction for new team members.
- Coordinated cross-functional data validation across major healthcare organisations, achieving 98% compliance and a 15% uplift in quality standards by aligning internal processes to regulatory requirements.

Picnic

Oct 2024 – present

Shopper (Part-time Werkstudent)

Mannheim, Germany

- Operational role alongside studies; managing time-sensitive, high-accuracy fulfilment tasks within a data-driven logistics environment.

PROJECTS

Real-Time Air Quality Monitoring System (AQMS)

Python, Zephyr RTOS, Nordic nRF52840, Streamlit, ASHRAE/WHO compliance

- Designed a 6-stage ETL pipeline processing multi-sensor environmental data with 99.8% packet integrity and a mean end-to-end latency of 29.7 ms, deployed on a Nordic nRF52840 microcontroller running Zephyr RTOS.
- Implemented regulatory compliance classification aligned with ASHRAE 62.1 and WHO standards, automatically categorising fused sensor readings into four severity tiers (OK → CRITICAL) for real-time health advisories.
- Built a Streamlit analytics dashboard delivering sub-second visual updates for non-technical users; authored full system documentation covering architecture, data flows, and sensor specifications.

ATM Transaction Performance Dashboard

Power BI, DAX, SQL

- Built a dynamic Power BI dashboard analysing ATM performance across monthly/yearly revenue, cost breakdown (maintenance vs. operational), uptime %, gross profit, and transaction trends segmented by ATM ID.

- Automated month-over-month comparative analysis using DAX time intelligence functions and custom KPI measures, reducing manual analysis time by 30% and improving reporting accuracy for operations teams.
- Designed interactive visualisations (bar graphs, area charts, slicers) optimised for non-technical stakeholders without SQL or BI tool experience.

Safety Monitoring System – Computer Vision (In Progress)

Python, YOLOv8n, OpenCV, Kaggle (Dual T4 GPU), Raspberry Pi 5, IMX500

- Training a YOLOv8n object detection model for real-time safety monitoring, achieving mAP@50 of 0.758 across classes including fire, intruder, weapon, and person detection.
- Targeting edge deployment on Raspberry Pi 5 with Sony IMX500 AI camera; optimising model size and inference latency for resource-constrained hardware without cloud dependency.
- Conducting training runs on Kaggle with dual T4 GPUs; managing multi-source dataset curation and class balancing for low-frequency hazard classes.

EDUCATION

Frankfurt University of Applied Sciences

MSc, High Integrity Systems (AI & Data Science)

Relevant coursework: Formal Software Verification, Smart Sensor Network Systems, AI & Machine Learning, Data Science Methods

Oct 2024 – present

Frankfurt am Main, Germany

University of Mumbai

BSc, Information Technology

2019 – 2022

Mumbai, India

ACHIEVEMENTS

- Best Performer of Q3 2023, eClinicalWorks – recognised for exceptional issue resolution speed and documentation quality across a high-volume support quarter.